Gendered Experiences of Ethnic Victims of Stalin's Repressions: Emotional Analysis of Oral Histories from the Gulag

Iashchenko, Iuliia
iuliia.iashchenko@uniroma1.it
La Sapienza University of Rome, Italy

Carteny, Andrea
andrea.carteny@uniroma1.it
La Sapienza University of Rome, Italy

Iashchenko, Anatolii
anatolii.iashchenko@uniroma1.it
La Sapienza University of Rome, Italy

Preserving Ukrainian's Cultural Heritage: From Photogrammetry to 3D Modeling for Post-War Restoration

Introduction

Odessa, a UNESCO World Heritage Site and a symbol of Ukraine's cultural legacy, has suffered significant damage during the ongoing Russo-Ukrainian war. This project proposes a systematic approach to preserving Odessa's cultural heritage through advanced photogrammetry and 3D modeling techniques. By documenting and reconstructing its historic architecture, the initiative ensures accurate restoration, safeguards cultural identity, and aids in post-war recovery efforts. In light of the destruction faced by several UNESCO-protected sites in Ukraine, the project leverages cutting-edge technology to protect and digitally restore Odessa's invaluable heritage.

Relevance

Odessa's historic center, renowned for its architectural blend of Baroque, Neoclassical, and Art Nouveau styles, is an essential element of Ukraine's cultural identity. The ongoing war has inflicted significant damage to heritage sites, including those designated by UNESCO as global landmarks. Efforts to preserve these sites are critical not only to maintain cultural resilience but also to facilitate the city's recovery after the war. In this context, 3D digitization has emerged as an essential tool for documenting, analyzing, and reconstructing these structures with precision.

Current Progress

The initial steps in creating a photogrammetry-based model of Odessa's damaged cathedral have been completed. This model is built from hundreds of images collected from museums, social media, and private contributors. Using Agisoft Metashape, these images were processed into a preliminary 3D representation. The project also awaits additional materials from Ukrainian state archives, including historical building plans, to refine and enhance the accuracy of the models. These initial efforts form the foundation for further work on other UNESCO-protected structures and threatened landmarks in Odessa.

Objectives

This project aims to:
1. Capture high-resolution imagery of Odessa's endangered heritage sites using photogrammetry.
2. Develop comprehensive 3D models as a basis for future restoration and virtual preservation.
3. Integrate archival data and historical documentation into the reconstruction process to ensure authenticity.
4. Foster interdisciplinary collaboration between historians, architects, and digital humanities specialists to create sustainable preservation methodologies.

Methodology

Data Collection
• Photogrammetry Tools:
• High-resolution cameras are employed to capture detailed structural images.
• Drones are utilized for aerial photography, enabling the documentation of large or inaccessible areas.
• Stabilizers ensure the accuracy and consistency of the captured data.
• Integration with Archival Data:
• Historical photographs, building plans, and postcards sourced from archives and libraries provide a comprehensive reference base.

• Satellite imagery establishes the condition of Odessa's buildings before the onset of the war, contributing to accurate pre-war reconstructions.
• Documents from Ukrainian state archives further enhance the authenticity and precision of the 3D models.

3D Modeling

• Software:
• Autodesk Revit and SketchUp are used for architectural modeling, while Blender and 3ds Max refine details and add textures.
• ZBrush enhances sections of buildings that are partially ruined, allowing for virtual restoration.
• Post-Processing:
• Unity and Unreal Engine are employed to create immersive virtual representations of the reconstructed sites.
• CloudCompare is used to refine point cloud data, ensuring high-quality digital replicas of the heritage sites.

Significance of 3D Modeling in Odessa's Preservation

Photogrammetry captures spatial data with high accuracy, but 3D modeling allows for deeper restoration and reconstruction efforts. These models offer the ability to address gaps or damage in structures by drawing upon historical references and archival material. As a result, they serve a dual purpose: as detailed blueprints for physical restoration and as digital archives for educational, cultural, and future recovery initiatives.

Cultural Significance of Odessa's Heritage

Odessa's architectural heritage represents a critical aspect of Ukraine's national identity, encompassing a diverse history and cultural richness that reflect the city's resilience and enduring spirit. As the war threatens to erase these symbols, digital preservation becomes an essential safeguard. Through 3D modeling, this project ensures that Odessa's cultural legacy is protected for future generations, even in the face of ongoing conflict.

Impact

• Preventing Irreversible Loss:

Immediate documentation through 3D modeling ensures that detailed records of Odessa's heritage are preserved, making future restoration efforts feasible.

• Rebuilding Ukrainian Cities:

The 3D models will provide the foundation for post-war restoration, not only in Odessa but across Ukraine, contributing to the rebuilding of urban centers and cultural renewal.

• Global Collaboration:

By involving international experts in Digital Humanities, this project highlights Odessa's cultural significance on a global scale, fostering collaboration and shared responsibility for preserving heritage.

Cultural Heritage Reports and Digital Tools

In recent reports by UNESCO, the devastation of Ukrainian cultural heritage has been emphasized, especially regarding the destruction of sites with immense historical value. These reports highlight the strategic importance of utilizing advanced digital tools for preservation, which aligns with the project's approach to documenting, restoring, and safeguarding Odessa's historical buildings. 3D modeling offers a means to digitally safeguard these sites, ensuring that their cultural importance is recognized worldwide. Moreover, photogrammetry has gained significant traction in the field of cultural heritage as a precise and efficient tool for capturing detailed information about historical structures.

Conclusion

Preserving Odessa's cultural heritage through 3D modeling is not just a technical achievement—it is an act of cultural defense and resilience. By combining photogrammetry with advanced digital modeling techniques, this project ensures the survival of Odessa's architectural legacy. These efforts provide both a foundation for post-war restoration and a digital archive for educational and cultural purposes. Through this work, Odessa's story of resilience will inspire future generations while contributing to the recovery of Ukrainian cities.

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